



Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from **3,900 transactions** to guide strategic business decisions.

Dataset at a Glance

3,900

Purchases

18

Columns

37

Missing Ratings

Key Features

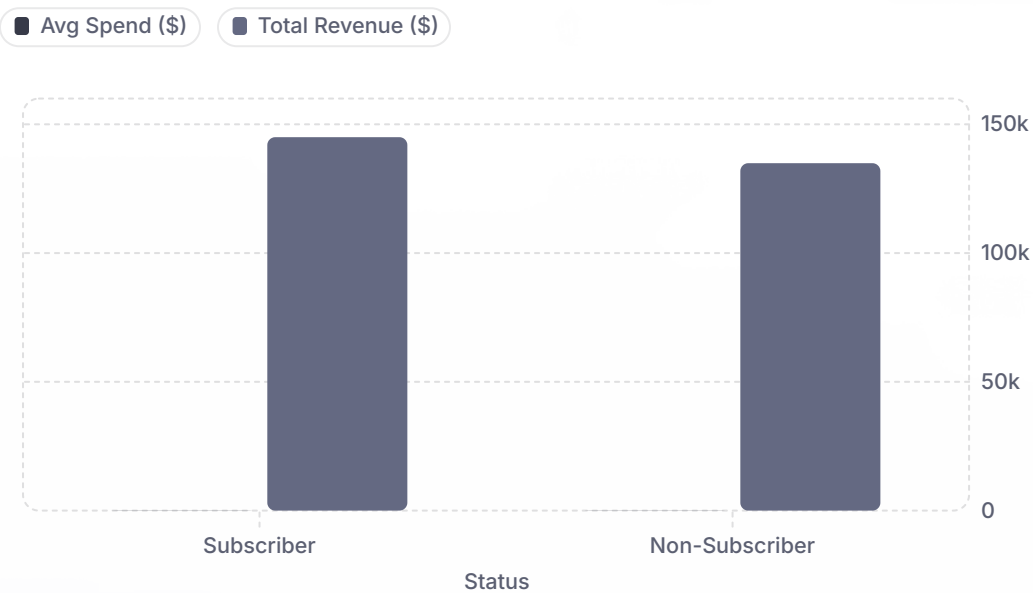
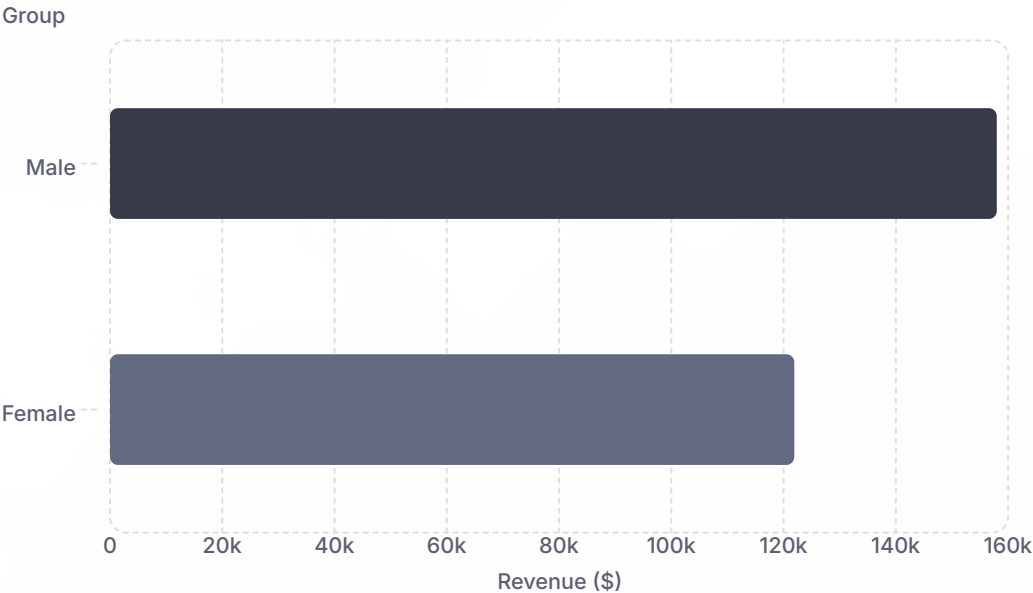
- **Demographics:** Age, Gender, Location, Subscription
- **Purchase:** Item, Category, Amount, Season, Size, Color
- **Behavior:** Discounts, Promo Codes, Frequency, Ratings, Shipping

Exploratory Data Analysis — Python



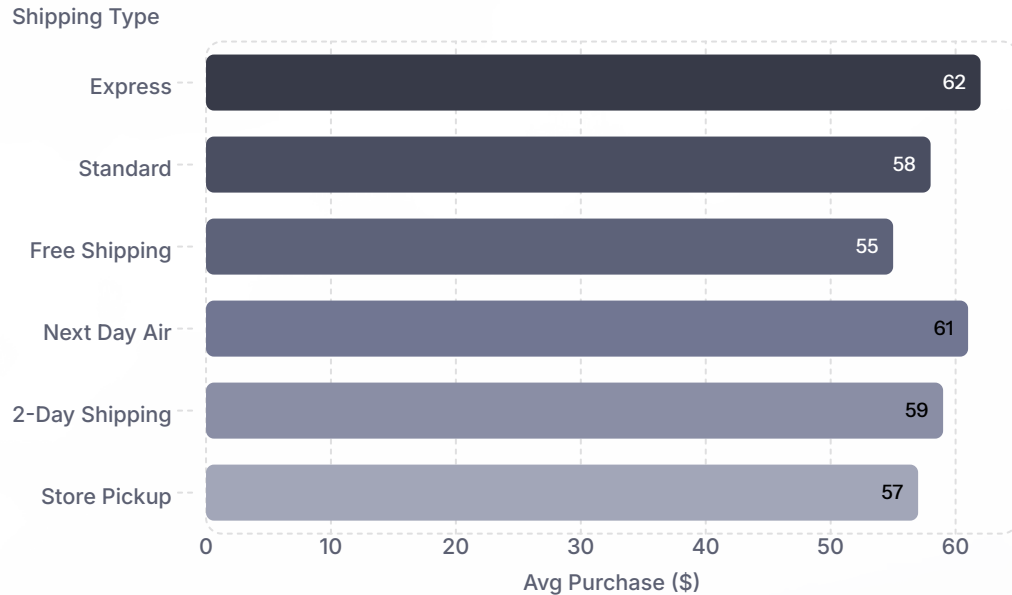
Column names standardized to snake_case; redundant `promo_code_used` dropped after confirming overlap with `discount_applied`.

Revenue by Gender & Subscription



Male customers generate higher total revenue. Subscribers spend slightly more on average and contribute greater total revenue.

Shipping Type & High-Spending Discount Users



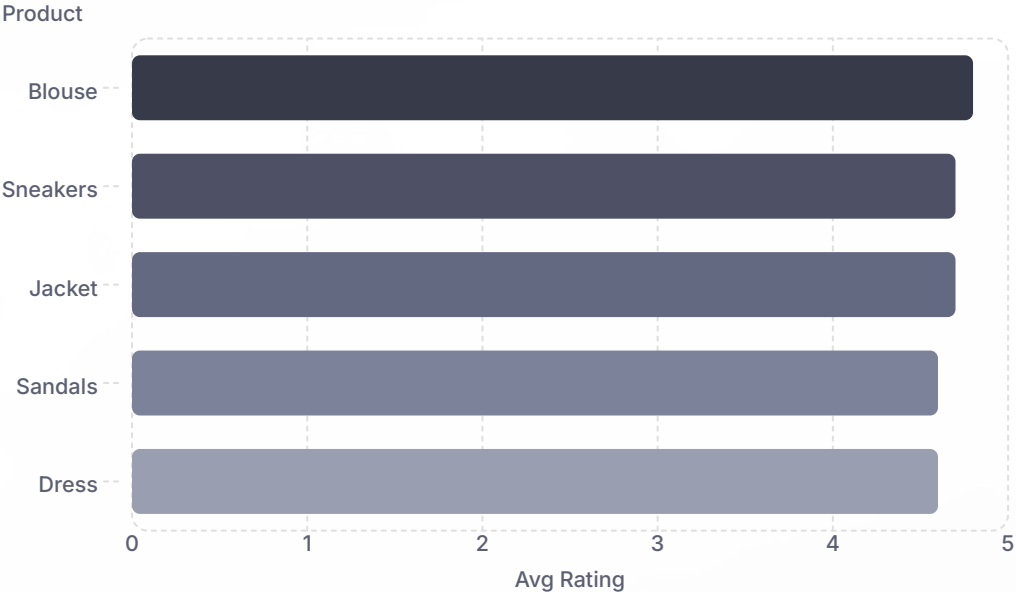
High-Spending Discount Users

A notable segment used discounts yet spent **above the average purchase amount** — indicating discount-resilient high-value customers worth targeting with loyalty perks.

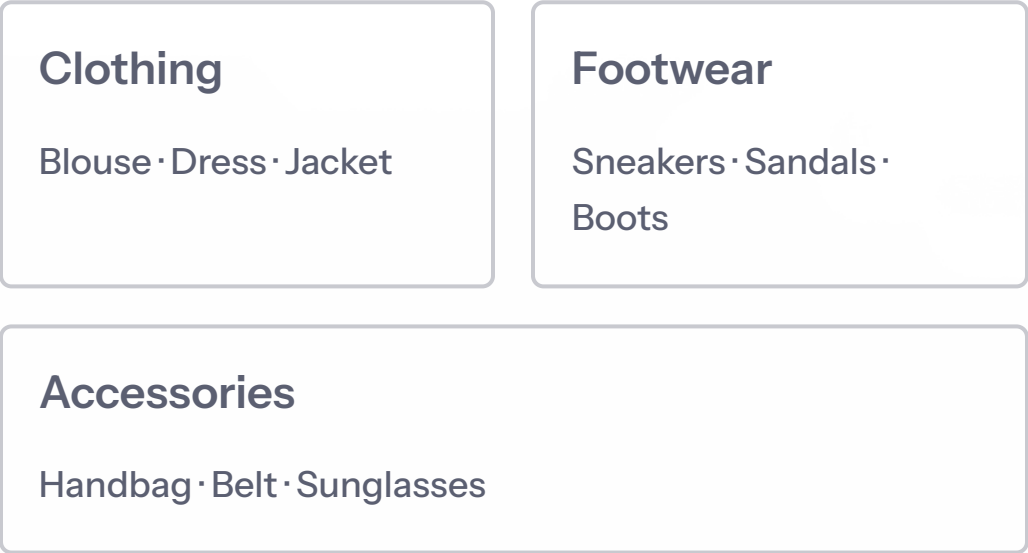
i Express shipping users show the highest average spend — a key segment for premium upsells.

Top Products by Rating & Category

Top 5 Products by Avg Rating



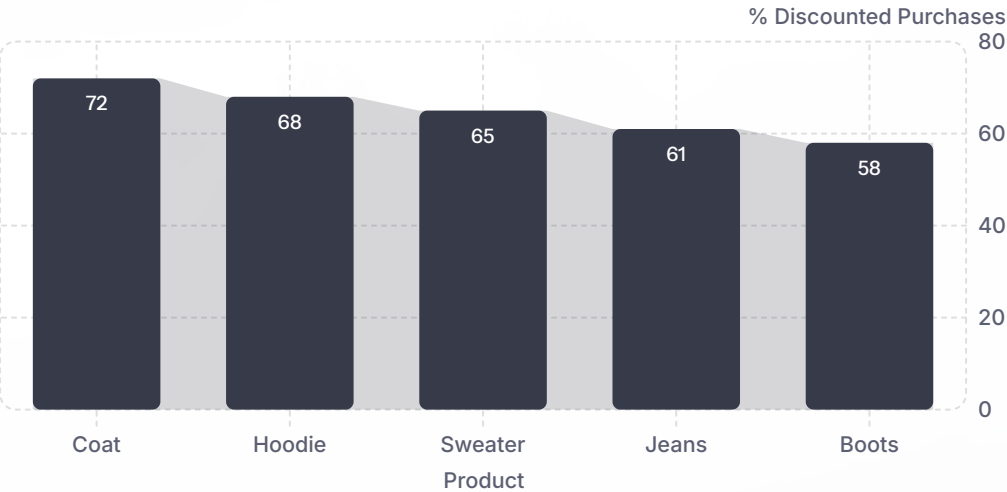
Top 3 Products per Category



Discount Dependency

80.56%

Total conversion rate

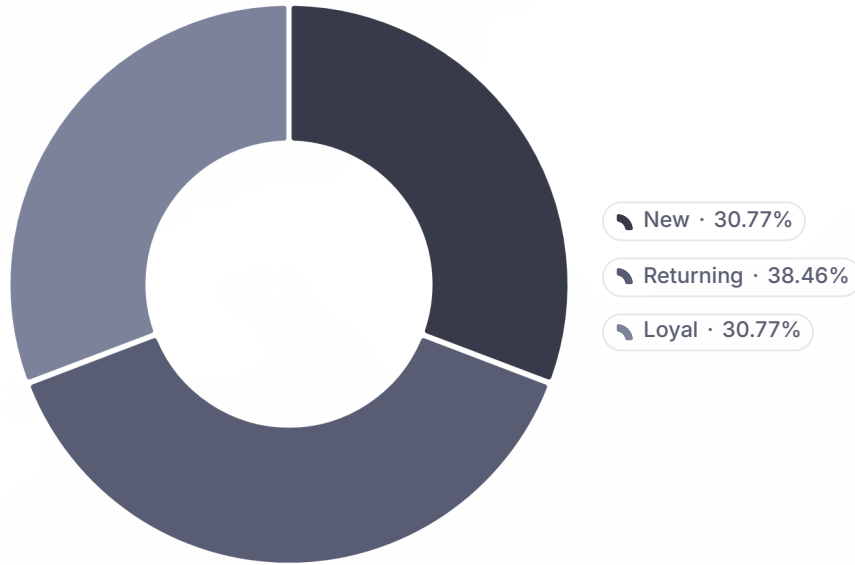


5 Most Discount-Dependent Products

These items see the **highest share of discounted purchases**, signaling potential margin risk if discounts are over-relied upon.

 Review discount policy for these products to protect profit margins.

Customer Segmentation & Loyalty



Repeat Buyers & Subscriptions

Customers with **more than 5 purchases** are significantly more likely to hold an active subscription — confirming that loyalty drives subscription uptake.

New

1–2 purchases

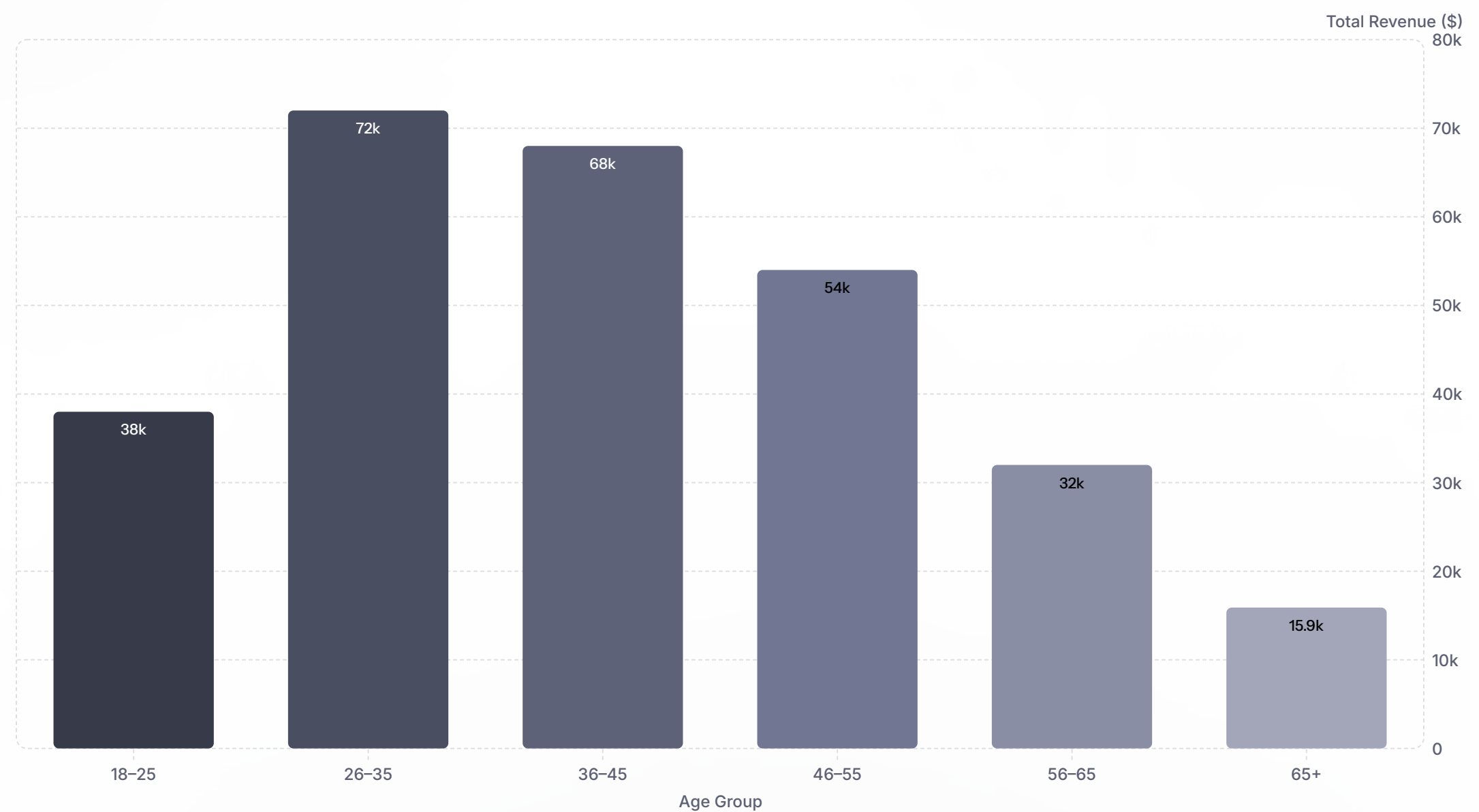
Returning

3–5 purchases

Loyal

5+ purchases

Revenue by Age Group



The **26-45 age group** drives the majority of revenue — the primary target for marketing investment and loyalty programs.

Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers → move to Loyal segment



Review Discount Policy

Balance sales boosts with margin control



Targeted Marketing

Focus on ages 26–45 and express-shipping users



Product Positioning

Highlight top-rated items in campaigns